Astronomy A3: Relativity and Black Holes

MAIS Physics & Astronomy Club (PAC)

The MAIS Physics & Astronomy Club crest shows a dark shadow encircled by a warped, glowing golden disk against the night sky. Far from being purely artistic, every feature of that image—the event horizon, the photon ring, and the lensed far side of the accretion disk—follows directly from gravity and the finite speed of light.

1. Escape Speed and the Event Horizon

In 1783, John Michell asked how compact a star of mass $\mathbf{M}$ would have to be for its Newtonian escape speed $\mathbf{v}_{esc} = \sqrt{(2\mathbf{GM}/\mathbf{R})}$ to reach the speed of light $\mathbf{c} = 3.00 \times 10^8$ m/s. Setting $\mathbf{v}_{esc} = \mathbf{c}$ yields the **Schwarzschild radius**:

$$\mathbf{R_s} = (2\mathbf{GM})/(\mathbf{c}^2) \approx 2.95 \text{ km } ((\mathbf{M})/(\mathbf{M}_\odot)). \tag{1}$$

Remark. Getting the exact prefactor 2 from (1)/ (2)$\mathbf{mc}^2 = \mathbf{GMm}/\mathbf{R}$ is a famous cancellation of two errors! Newtonian mechanics uses neither relativistic kinetic energy nor curved spacetime; in Einstein's general relativity (solved by Karl Schwarzschild in 1916), light does not "slow down and fall back"—rather, inside $\mathbf{r} = \mathbf{R_s}$ spacetime is curved so strongly that all future-directed paths point toward smaller $\mathbf{r}$.

[1] Problem 1. Using equation (1) with $\mathbf{G} = 6.67 \times 10^{-11}$ N·m^2/kg^2 and $\mathbf{c} = 3.00 \times 10^8$ m/s:

(a) Compute the Schwarzschild radius $\mathbf{R_s}$ of Earth ($\mathbf{M}_\oplus = 5.97 \times 10^{24}$ kg) in millimeters, and of the Milky Way's central black hole Sagittarius A* ($\mathbf{M} \approx 4.15 \times 10^6\ \mathbf{M}_\odot$) in kilometers and in AU.

(b) Suppose the Sun were instantaneously replaced by a 1 $\mathbf{M}_\odot$ black hole without losing any mass. What would happen to Earth's orbit?

2. Tidal Forces Across Extended Bodies

In free fall, an observer feels weightless because every part of their body accelerates together—**except** for the slight variation of gravity from head to toe! For a radial rod of small length $\ell \ll \mathbf{r}$ at distance $\mathbf{r}$ from mass $\mathbf{M}$, differentiating $\mathbf{g}(\mathbf{r}) = \mathbf{GM}/\mathbf{r}^2$ gives the stretching tidal acceleration between its ends:

$$\Delta\mathbf{a} = \mathbf{g}(\mathbf{r} - (\ell)/(2)) - \mathbf{g}(\mathbf{r} + (\ell)/(2)) \approx (2\mathbf{GM}\ell)/(\mathbf{r}^3). \tag{2}$$

[2] Problem 2 (Stellar vs. supermassive horizons). Define the effective "horizon density" $\bar{\rho}_s = \mathbf{M} / ((4)/(3)\pi \mathbf{R_s}^3)$, and consider an astronaut of height $\ell = 1.8$ m falling feet-first through the horizon at $\mathbf{r} = \mathbf{R_s}$.

(a) Show that both $\bar{\rho}_s$ and the horizon tidal acceleration $\Delta\mathbf{a}(\mathbf{R_s})$ scale as $\mathbf{M}^{-2}$—getting **weaker** as the black hole gets heavier!

(b) Calculate $\Delta\mathbf{a}(\mathbf{R_s})$ (in units of Earth gravity $\mathbf{g}_0 = 9.8$ m/s^2) for a stellar-mass black hole of 10 $\mathbf{M}_\odot$ versus a supermassive black hole of 10^8 $\mathbf{M}_\odot$, and compute $\bar{\rho}_s$ for the 10^8 $\mathbf{M}_\odot$ black hole.

3. Accretion Disks and Gravitational Lensing

Isolated black holes are completely dark, yet accreting black holes power the brightest quasars in the universe. Infalling gas carries angular momentum and flattens into a hot **accretion disk**; viscous friction causes gas to spiral inward down to the innermost stable circular orbit ($\mathbf{r}_{\mathrm{ISCO}} = 3\mathbf{R_s}$ for a non-spinning black hole), radiating away gravitational binding energy before plunging across the horizon. Meanwhile, light passing near the photon sphere ($\mathbf{r}_{\mathrm{ph}} = 1.5\mathbf{R_s}$) bends so strongly that rays emitted from the **top and bottom of the far side** of the disk curve over and under the black hole toward our eyes—producing the double-arched halo seen on our club crest!

[2] Problem 3 (Why quasars outshine galaxies). Estimate the energy released when a parcel of gas of mass $\mathbf{m}$ spirals slowly from infinity down to a circular orbit at $\mathbf{r}_{\mathrm{ISCO}} = 3\mathbf{R_s} = 6\mathbf{GM}/\mathbf{c}^2$.

(a) Using the Newtonian circular-orbit energy $\mathbf{E} = -\mathbf{GMm}/(2\mathbf{r})$ as a simple estimate, find the radiated energy as a fraction of rest-mass energy $\mathbf{mc}^2$, and compare it with the 0.7% efficiency of hydrogen nuclear fusion in stars. (Exact general relativity gives $1 - \sqrt{(8/9)} \approx 5.7\%$ for a non-spinning black hole, and up to 42% for a rapidly spinning black hole!)

(b) Using the relativistic efficiency $\eta = 0.10$ (10%), what mass accretion rate $\mathbf{dM}/\mathbf{dt}$ (in $\mathbf{M}_\odot$ per year) is needed to power a luminous quasar shining at $\mathbf{L} = 10^{13}$ $\mathbf{L}_\odot$ ($\mathbf{L}_\odot = 3.83 \times 10^{26}$ W)?

[3] Problem 4 (Weighing Sagittarius A* with the star S2). Over three decades of infrared adaptive-optics observations, astronomers tracked the B-type star S2 completing full elliptical orbits around the compact radio source Sagittarius A* at the center of our Milky Way (distance $\mathbf{d} = 8{,}200$ pc). S2's orbit has period $\mathbf{P} = 16.05$ yr, eccentricity $\mathbf{e} = 0.885$, and apparent angular semi-major axis $\theta_{\mathbf{a}} = 0.125''$.

(a) Convert $\theta_{\mathbf{a}}$ at distance $\mathbf{d} = 8{,}200$ pc into S2's physical semi-major axis $\mathbf{a}$ in AU, and use Kepler's third law to determine the central mass $\mathbf{M}$ in units of $\mathbf{M}_\odot$.

(b) Find S2's closest approach distance $\mathbf{r_p} = \mathbf{a}(1 - \mathbf{e})$ in AU, and use the vis-viva equation to compute S2's speed at pericenter as a percentage of the speed of light $\mathbf{c}$.